

Message bits $\rightarrow$ Tx 1011 $\xrightarrow{\text{wireless channel}}$ Rx 1111 (Noise)

$\left[\begin{array}{c} \text{more 0's} \\ \text{less 1's} \end{array} \right]$

Error correction (Theoretical)

LDPC (Low Density Parity Check) $\rightarrow$ adds extra bits $\rightarrow$ (sparse parity check matrix)

dense matrix

$$H = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}$$

to fix errors using iterative decoding

sparse $\rightarrow$ spread out (0's, 1's)

$$H = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

A	B	A ^ B
0	0	0
0	1	1
1	0	1
1	1	0

Shannon Capacity

$$C = B \log_2(1 + \text{SNR})$$

$$B = 1$$

$$R = 1.08$$

$R \leq C$ (reliable transmission)

$$1.08 \leq \log_2(1 + \text{SNR})$$

$$2^{1.08} \leq 1 + \text{SNR}$$

$$3.48 \leq 1 + \text{SNR}$$

$$\text{SNR} \geq 2.48$$

$$\boxed{\text{SNR}_{\min} = 2.48}$$

$$\boxed{2^{1.08} \approx 3.48}$$

Transmission rate $\leq$ channel capacity

$$p_1 = m_1 \wedge m_2 = 1 \wedge 0 = 1$$

$$p_2 = m_2 \wedge m_3 = 0 \wedge 1 = 1$$

$$p_3 = m_1 \wedge m_3 = 1 \wedge 1 = 0$$

$$C = [101110]$$

Now

$$m = [101] \Rightarrow k = 3$$

Add 3 parity bits $\rightarrow$ total 6 bits

$$C = [101 p_1 p_2 p_3] \Rightarrow n = 6$$

$$\boxed{R = 0.5}$$

$$R = \frac{k}{n} = \frac{3}{6}$$

code rate

$$0.5 \leq \log_2(1 + \text{SNR})$$

$$2^{0.5} \leq 1 + \text{SNR}$$

$$1.414 \leq 1 + \text{SNR}$$

$$\text{SNR} \geq 0.414$$

$$\boxed{\text{SNR}_{\min} = 0.414} \Rightarrow \text{theoretical SNR}$$

dB

$$10 \log_{10}(0.414) \approx -3.83 \text{ dB} \rightarrow \text{needed to transmit at rate } 0.5$$

$\rightarrow$ checks for error multiple times and correct it.

(Practical)

1) $R = 0.5$ (Fix code Rate)

$$R = \frac{k}{n}$$

2) Generate data

Ex: 100000 random bits

3) Encode

$$m = [101] \quad n = 6 \quad c = [101110]$$

4) Pass through channel :- $y = kx + n$

$x \rightarrow$ transmitted encoded bits

$n \rightarrow$ gaussian noise $n \sim N(0, \sigma^2)$

5) Decode :- Noisy y

LDPC $\rightarrow$ Belief propagation

Iteratively

updates probability of each bit as 0 (or) 1

To correct the bits it uses

$\rightarrow$ parity check (H matrix)

Output $\rightarrow \hat{m} \quad (101)$

6) BER

Ex: 1000 errors $\rightarrow$ 100,000 bits $\rightarrow$ BER = 0.01 = 10^{-2}

SNR (dB)

BER

-4

10^{-2}

-3.8

10^{-3}

-3.2

10^{-4}

-3

10^{-5}

SNR $\uparrow \rightarrow$ decoding becomes reliable

LDPC achieves BER $\leq 10^{-5}$ at -3 dB

Shannon SNR_{min} $R = 0.5 =$ -3.83 dB

(Near-Shannon performance)

diff
} 0.83